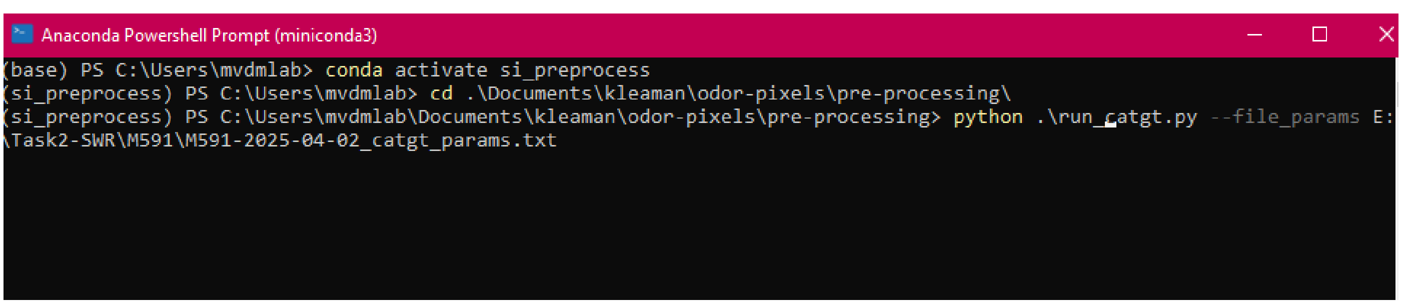
Neuropixels Prepro Pipeline
This preprocessing pipeline assumes you are starting with raw neuropixels data files acquired by SpikeGLX.

Setup
1. Clone the odorpixels repo from the vandermeerlab github
2. Follow the steps in 'odor-pixels/pre-processing/EnvironmentSetup.md' to create the necessary conda environment
3. Download CatGT and TPrime from SpikeGLX
4. Install phy2 (follow developer instructions to create phy conda environment)
5. Download npy-matlab

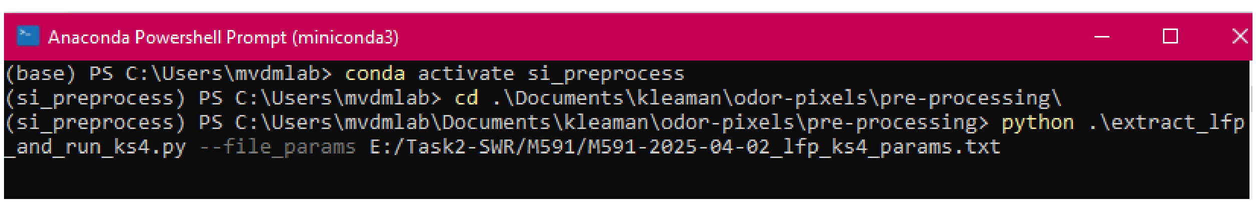
Preprocessing Steps
1. **Run CatGT** (Temporally aligns channels within each probe & extracts TTL pulses)
    a. Copy file 'odor-pixels/pre-processing/example_catgt_params.txt' for each session to preprocess, paste in subject folder & rename
    b. Fill in required parameters in newly copied catgt_params file:
        i. SOURCE_DIR indicates parent directory that contains the raw data folder for the session
        ii. DEST_DIR indicates directory you want catgt folder to be output. **Ideally want this to output to a fast SSD.
        iii. CATGT_PATH is the path to the installed catgt tool
        iv. SESSION_NAME should be the name of the raw data session folder preceding the suffix "_g0". Typically formatted as MouseID_Date (e.g., M591-2025-04-02).
        v. PROBE_INDICES = 0 for one probe, = 0,1 for two probes
        vi. Adjust analog and digital channels based on which channels contain task information from the NIDAQ
    c. In Anaconda PowerShell, activate conda environment si_preprocess, cd to 'odor-pixels/preprocessing', and run script "run_catgt.py". Set file-params to the path to the catgt_params.txt file created above.
    d. Example call:



    e. To run catgt on multiple sessions together, make catgt_params files for all desired sessions. Then make a copy of the file 'odor-pixels/pre-processing/example_batch_params.txt' and paste paths to the catgt_params files for all sessions into the batch_params file.

f. Call 'run_catgt.py' as shown above but replace "`--file_params`" with "`--batch_params`" and the path to the batch_params.txt file

2. **Run Kilosort & Extract LFP** (Kilosort will automatically cluster and label putative spiking units. LFP signal will be extracted from a subset of channels on a probe and saved in .mat files that can be loaded as mvdmlab datatypes. These are separate steps that are run through a combined script)
    a. Copy file 'odor-pixels/pre-processing/example_lfp_ks4_params.txt' for each session to preprocess, paste in subject folder & rename
    b. Fill in parameters:
        i. CATGT_OUTPUT_DIR: path to catgt folder for session (will be formatted 'catgt_[session_name]_g0')
        ii. OUTPUT_DIR: Where LFP files should be output. Typically a subfolder of the subject directory called 'preprocessed'
        iii. SESSION_ID: Session name, typically formatted MouseID_Date
        iv. FAST_STORAGE_DIR: Path to a fast SSD that Kilosort temporary files will write to
        v. PROBE_INDICES = 0 for one probe, = 0,1 for two probes
        vi. EXTRACT_LFP_FOR: set to 0 and/or 1 to process specific probes. Leave blank if you don't want to extract LFP.
        vii. RUN_KILOSORT_FOR: set to 0 and/or 1 to process specific probes. Leave blank if you don't want to run KS.
    c. In Anaconda PowerShell, activate conda environment si_preprocess, cd to 'odor-pixels/preprocessing', and run script "extract_lfp_and_run_ks4.py". Set file-params to the path to the lfp_ks4_params.txt file created above.
    d. Example call:



```
Anaconda Powershell Prompt (miniconda3)
(base) PS C:\Users\mvdmlab> conda activate si_preprocess
(si_preprocess) PS C:\Users\mvdmlab> cd .\Documents\kleaman\odor-pixels\pre-processing\
(si_preprocess) PS C:\Users\mvdmlab\Documents\kleaman\odor-pixels\pre-processing> python .\extract_lfp_and_run_ks4.py --file_params E:/Task2-SWR/M591/M591-2025-04-02_lfp_ks4_params.txt
```

    e. To run ks4 and extract lfp for multiple sessions together, repeat batch_params instructions as explained before

3. **Run Extract Raw Analog Signals** (<u>Only applicable if raw analog signal exists.</u> Extracts timeseries data from specified analog channels in the NIDQ stream of a Neuropixels recording. For use with non-event/TTL related signals (e.g. fiber photometry data). Filters, downsamples, and saves signal as a .mat file)
    a. Copy file 'odor-pixels/pre-processing/example_raw_analog_params.txt' for each session to preprocess, paste in subject folder & rename
    b. Fill in parameters:

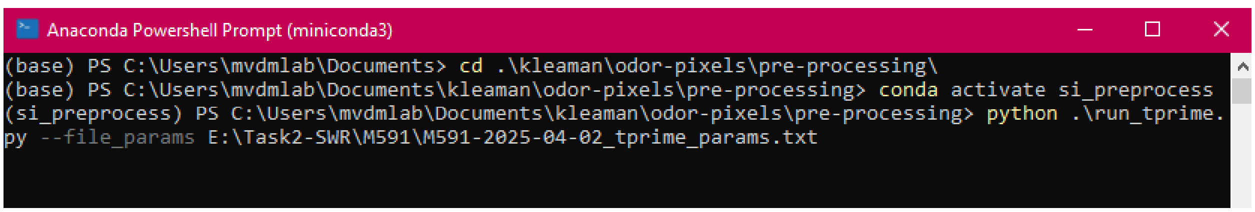
        i. SOURCE_DIR: Path to the original recording directory (contains .nidq.bin and .nidq.meta)
        ii. CATGT_OUTPUT_DIR: path to catgt folder for session (Where per-channel _raw_times.npy files for TPrime will be written)
        iii. OUTPUT_DIR: Typically a subfolder of the subject directory called 'preprocessed' followed by a session subfolder. Where per-channel .mat files will be saved
        iv. SESSION_ID: Session name, typically formatted MouseID_Date
        v. RAW_ANALOG_CHANNELS: JSON dict mapping channel name for analog channels to be extracted to preprocessing parameters (see param file for examples)
        vi. Global Defaults section: Default pre-processing applied to any channel whose list is missing a value (optionally editable).
            1. RAW_ANALOG_DEFAULT_HIGH_PASS = 0.1
            2. RAW_ANALOG_DEFAULT_LOW_PASS = 500.0
            3. RAW_ANALOG_DEFAULT_SAMPLE_RATE = 1000
    c. In Anaconda PowerShell, activate conda environment si_preprocess, cd to 'odor-pixels/preprocessing', and run script "extract_raw_analog_signals.py". Set file-params to the path to the raw_analog_params.txt file created above.
    d. Example call:

```
(base) PS C:\Users\mvdmlab> conda activate si_preprocess
(si_preprocess) PS C:\Users\mvdmlab> cd C:\Users\mvdmlab\Documents\egibbens\odor-pixels\pre-processing
(si_preprocess) PS C:\Users\mvdmlab\Documents\egibbens\odor-pixels\pre-processing>  python .\extract_raw_analog_signals.
py --file_params D:\vCA1\M653\M653_2026_02_03_raw_analog_params.txt
```

    e. To extract raw analog signals for multiple sessions together, repeat batch_params instructions as explained before

4. **Run TPrime** (TPrime synchronizes data from multiple probes and the NIDAQ by aligning the sync pulse from each data stream to the sync pulse on imec0)
    a. Copy file 'odor-pixels/pre-processing/example_tprime_params.txt' for each session to preprocess, paste in subject folder & rename
    b. Fill in parameters:
        i. CATGT_OUTPUT_DIR: path to catgt folder for session (will be formatted 'catgt_[session_name]_g0', e.g. catgt_M591-2025-04-02_g0)
        ii. OUTPUT_DIR: Path to folder where LFP files should be output. Typically a subfolder of the subject directory called 'preprocessed' followed by a session subfolder (e.g. E:\Task2-SWR\M591\preprocessed\M591-2025-04-02)
        iii. SESSION_ID: Session name, typically formatted MoueID_Date
        iv. TPRIME_PATH: Path to TPrime folder installed from SpikeGLX
        v. PROBE_INDICES = 0 for one probe, = 0,1 for two probes
        vi. TO_PROBE: Probe to which others will be aligned, generally 0
        vii. ALIGN_EVENTS, ALIGN_SPIKES, ALIGN_LFP: Set all to 'True' to align all data. Set any of these to 'False' in the event you don't want to align those data.

        viii. ALIGN_RAW_ANALOG: Set to 'True' if you ran extract_raw_analog_signals.py and want to align the extracted raw analog signals.
        ix. ANALOG_CHANNELS, DIGITAL_CHANNELS: List channels present in this recording. Only list channels with non-neural TTL/event times that were extracted with CatGT (e.g. tcat.nidq. file exists).
        x. RAW_ANALOG_CHANNELS: List raw analog channels that you want to align (if applicable). These are the channels you extracted with extract_raw_analog_signals.py
        xi. KS_OUTPUT_FOLDERS: Paste path to the kilosort output folder for imec1 generated when KS4 is run in the previous step. Not applicable for single probe recordings. This path should end in a folder called "sorter_output".
    c. In Anaconda PowerShell, activate conda environment si_preprocess, cd to 'odor-pixels/preprocessing', and run script "run_tprime.py". Set file-params to the path to the tprime_params.txt file created above.
    d. Example call:

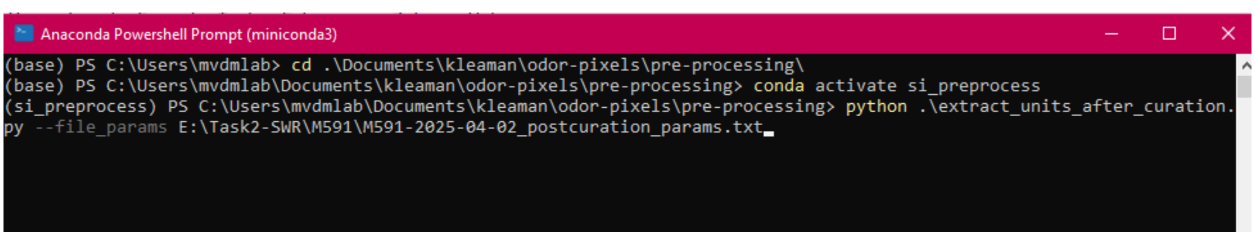


    e. To run tprime for multiple sessions together, repeat batch_params instructions as explained before

5. **Manually curate spiking data using phy** (the phy GUI displays the units identified by kilosort and allows the user to label units as good, MUA, or noise)
    a. In a new Anaconda Powershell window, cd to the kilosort output folder for the first probe from the desired session (ex. `cd F:/KS4_outputs/M591-2025-04-02_imec0/sorter_output`)
    b. Activate phy conda environment (may be called "phy2" if using updated installation instructions)
    c. Run the command `phy extract-waveforms params.py` to extract waveforms from the kilosort output for GUI display. This step only ever run once (before the first time you go to manually curate units).
    d. Run the command `phy template-gui params.py` to launch the phy GUI
        i. For detailed instructions / tips for manual curation using phy, there is a document called "Manual Curation with Phy" in the vandermeerlab github repo mvdmlab-protocols/neuropixels. Also see the phy developer documentation and practical guide [here](#).

e. Save (using the save symbol in the GUI) frequently as phy can crash. Close the GUI window after saving once all clusters have been curated. Repeat this process if there is a second probe in the session.

6. **Extract curated units** (this process extracts spike times, waveform information, etc from the phy output and saves them as .mat files that can be loaded as mvdmlab datatypes)
   a. Copy file 'odor-pixels/pre-processing/example_extract_postcuration_params.txt' for each session to preprocess, paste in subject folder & rename
   b. Fill in parameters:
      i. CATGT_OUTPUT_DIR: path to catgt folder for session (will be formatted 'catgt_[session_name]_g0', e.g. catgt_M591-2025-04-02_g0)
      ii. OUTPUT_DIR: Path to folder where LFP files should be output. Typically a subfolder of the subject directory called 'preprocessed' followed by a session subfolder (e.g. E:\Task2-SWR\M591\preprocessed\M591-2025-04-02)
      iii. SESSION_ID: Session name, typically formatted MoueID_Date
      iv. PROBE_INDICES = 0 for one probe, = 0,1 for two probes
      v. SORTER_OUTPUT_FOLDERS: Paths to the kilosort/phy output directory formatted with double slashes to be python compatible (example: `F://KS4_outputs//M591-2025-04-02_imec0//sorter_output`)
      vi. PREPROCESS_FOLDERS: Path to folders appended with imec0_si_preprocess and imec1_si_preprocess which are output as temporary files by kilosort into the same parent directory as the sorter_output_folders listed above.
      vii. SAVE_WAVEFORMS, EXTRACT_GOOD_UNITS, EXTRACT_MUA_UNITS: Can be set to True/False based on what information you want to extract. Typically, all are set to True.
   c. In Anaconda PowerShell, activate conda environment si_preprocess, cd to 'odor-pixels/preprocessing', and run script "extract_units_after_curation.py". Set file-params to the path to the postcuration_params.txt file created above.
   d. Example call:



   e. To extract units for multiple sessions together, repeat batch_params instructions as explained before

7. **Sync LFP for multiple probes** (running TPrime previously should have generated a file in the catgt output folder called imec1_adj_lfp_times.npy, which will allow the timepoints

in the imec1 LFP.mat file to be aligned with the imec0 LFP times. Must be run regardless of whether the recording had 1 or 2 probes because of data structure reformatting that occurs)

   a. Launch the file 'syncProbeLFP.m' in matlab
   b. Paste the path to the preprocessed folder that contains the lfp.mat files for this session into the cd() field
   c. Paste the path to the imec1_adj_lfp_times.npy file for this session in the ReadNPY() file
   d. Run in matlab
   e. The script will save files 'imec1_clean_lfp_new.mat' and 'imec0_clean_lfp_new.mat'. For compatibility with analysis scripts, the old imec0 and imec1 lfp files should be deleted or renamed, and these new scripts should be saved as 'imec1_clean_lfp.mat' and 'imec0_clean_lfp.mat'.

8. **Sync analog signal to probe** (<u>Only applicable if raw analog signal exists.</u> Running TPrime previously should have generated a file in the catgt output folder called {channel}_adj_raw_times.npy, which will allow the timepoints in the {channel}_unsynced.mat file to be aligned with the imec0 LFP times)
   a. Launch the file 'syncAnalogToProbe.m' in matlab
   b. channels: put the raw analog channels to process
   c. output_dir: path to the directory that contains the {channel}_unsynced.mat files (OUTPUT_DIR from params)
   d. catgt_output_dir: path to the directory containing the {channel}_adj_raw_times.npy files (CATGT_OUTPUT_DIR)
   e. Run in matlab.
   f. The script will save files {channel}_synced.mat in the output directory that will be used for later analysis. Old _unsynced.mat files may be deleted (optional).
   g. Example:

```
% —— User-configurable parameters ——————————————————————————————

addpath(genpath('D:\npy-matlab'))   % path to npy-matlab

% Channels to process (must match names used in the Python extraction scripts)
channels = {'XA7'};

% Directory containing the {channel}_raw.mat files (OUTPUT_DIR from params)
output_dir = 'D:\vCA1\M653\preprocessed\M653_2026_02_05';

% Directory containing the {channel}_adj_raw_times.npy files (CATGT_OUTPUT_DIR)
catgt_output_dir = 'D:\Temp\catgt_M653_2026_02_05_g0';
```

9. **Convert task events to .mat file** (converts XA_ON and XA_OFF text files output by tprime to a format compatible with mvdmlab codebase)
   a. Launch the file 'txtTOevt.m' in matlab
   b. Paste the filepaths for all preprocessed directories into the folder_list field
   c. Check to ensure that the correct channels are being added to the event object with the correct labels
   d. Run the file. It will output a .mat file containing the onset timepoints for all events.

10. **Ensure that the ExpKeys.mat file is complete**
    a. This file should contain information + metadata about the session and should be placed into the prepro folder

11. **Convert to .NWB** (This is technically optional but will make the preprocessed session easier to share & will ensure compatibility with some mvdmlab odorpixels analysis scripts. Necessary for decorrelation analysis, odor responsivity, odor selectivity, etc.)
    a. Clone the [mvdmlab_npx_to_nwb repository]
    b. Create new python environment called mvdmlab_nwbconverter from 'mvdmlab_npx_to_nwb/src/environment.yml'
    c. In a text editor, open 'mvdmlab_npx_to_nwb/src/convert_single_session.py' (to convert one session) or 'mvdmlab_npx_to_nwb/src/convert_multiple_sessions.py' to convert additional sessions together
        i. In convert_single_session.py, paste the preprocessed session filepath into input_dir (formatted with double slashes to be python compatible) and paste the desired output filepath below appended with the desired .nwb file name
        ii. In convert_multiple_sessions.py, paste filepaths to all preprocessed session folders you want to convert.
    d. In an Anaconda Powershell prompt, activate the mvdmlab_nwbconverter python environment and call the desired conversion file to convert one / multiple sessions

Documented issues / tips:

- **Tprime with only 1 probe (no longer applicable as of 5/4/26):** Some odd errors can occur when running the prepro scripts on a session with only one probe, as many scripts were designed for primary use with dual probe recording sessions. This is particularly an issue with TPrime, because there is no imec1 probe to align with the imec0 times, but you still need to extract the analog and digital events and align those with the imec0 sync pulses. This script may be fixed in the future, but at present this requires manually calling TPrime directly from Windows Powershell. That call looks like this:

```
Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\mvdmlab> D:\TPrime-win/TPrime.exe -syncperiod='1.0' -tostream='D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g
0_tcat.imec0.ap.xd_384_6_500.txt' -fromstream='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xd_8_0_500.txt
' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xa_0_0.txt,E:\Task2-SWR\M590\preprocessed\M590-202
5-04-05\XA0_ON.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xia_0_0.txt,E:\Task2-SWR\M590\pr
eprocessed\M590-2025-04-05\XA0_OFF.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xa_1_0.txt,E
:\Task2-SWR\M590\preprocessed\M590-2025-04-05\XA1_ON.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat
.nidq.xia_1_0.txt,E:\Task2-SWR\M590\preprocessed\M590-2025-04-05\XA1_OFF.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M59
0-2025-04-05_g0_tcat.nidq.xa_2_0.txt,E:\Task2-SWR\M590\preprocessed\M590-2025-04-05\XA2_ON.txt' -events='5,D:\catgt_M590
-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xia_2_0.txt,E:\Task2-SWR\M590\preprocessed\M590-2025-04-05\XA2_OFF.txt' -eve
nts='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xa_3_0.txt,E:\Task2-SWR\M590\preprocessed\M590-2025-04-0
5\XA3_ON.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xia_3_0.txt,E:\Task2-SWR\M590\preproce
ssed\M590-2025-04-05\XA3_OFF.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xa_4_0.txt,E:\Task
2-SWR\M590\preprocessed\M590-2025-04-05\XA4_ON.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.
xia_4_0.txt,E:\Task2-SWR\M590\preprocessed\M590-2025-04-05\XA4_OFF.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025
-04-05_g0_tcat.nidq.xa_5_0.txt,E:\Task2-SWR\M590\preprocessed\M590-2025-04-05\XA5_ON.txt' -events='5,D:\catgt_M590-2025-
04-05_g0\M590-2025-04-05_g0_tcat.nidq.xia_5_0.txt,E:\Task2-SWR\M590\preprocessed\M590-2025-04-05\XA5_OFF.txt' -events='5
,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xa_6_0.txt,E:\Task2-SWR\M590\preprocessed\M590-2025-04-05\XA6_
ON.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xia_6_0.txt,E:\Task2-SWR\M590\preprocessed\M
590-2025-04-05\XA6_OFF.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xa_7_0.txt,E:\Task2-SWR\
M590\preprocessed\M590-2025-04-05\XA7_ON.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xia_7_
0.txt,E:\Task2-SWR\M590\preprocessed\M590-2025-04-05\XA7_OFF.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05
_g0_tcat.nidq.xd_8_2_0.txt,E:\Task2-SWR\M590\preprocessed\M590-2025-04-05\XD2_ON.txt' -events='5,D:\catgt_M590-2025-04-0
5_g0\M590-2025-04-05_g0_tcat.nidq.xid_8_2_0.txt,E:\Task2-SWR\M590\preprocessed\M590-2025-04-05\XD2_OFF.txt' -events='5,D
:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xd_8_3_0.txt,E:\Task2-SWR\M590\preprocessed\M590-2025-04-05\XD3_
ON.txt' -events='5,D:\catgt_M590-2025-04-05_g0\M590-2025-04-05_g0_tcat.nidq.xid_8_3_0.txt,E:\Task2-SWR\M590\preprocessed
\M590-2025-04-05\XD3_OFF.txt'
```

  - This call is difficult to parse. It is easiest to generate this TPrime call using a jupyterlab notebook (either preProcess_Kyoko or preProcess_Manish.ipynb) in the odor-pixels/pre-processing/legacy folder. Scroll down to 'TPrime command generator to align events'. Edit the tprime_path, tsource_path, and tdest_path as needed (tsource should be the path to the catgt folder, tdest should be the preprocessed folder)
  - Run the subsequent sections until the final command is printed. This can be pasted into the windows powershell window and run to extract and align events.

- **Different session end timepoints between imec0 and imec1:** If SpikeGLX experienced an issue during the recording and ended the acquisition early, there can be cases where the length of imec0 (as in, the final timepoint in the session) is different than the length of imec1 and nidq.
  - In these cases, you can manually check the interval of the sync pulses on imec0, imec1, and the nidq after extracting these timepoints with catgt. The files with the sync pulse edge times are denoted as _500.txt files. If there are missing timepoints in the middle of the recording, you will see a jump somewhere in the list of timepoints; otherwise, the interval between each subsequent sync pulse

edge is ~1 sec. Because in the case of these sessions the sync pulse times are completely uniform (1 sec intervals) but the sync pulse times for imec0 just end 100 sec earlier, it indicates the problem only occurs at the end of the session (meaning the earlier timepoints of the session are still usable & the two probes can still be synchronized with each other successfully).

- To solve this mismatch in final timepoints, there is an optional catgt field (maxsecs) which can be used to only extract data prior to a specific timepoint. This way both probes and the nidq data can be extracted only prior to the time where imec0 failed, allowing all events to be correctly synchronized and making the sessions still usable.
- Example call:

```
Windows PowerShell

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\mvdmlab> C:\CatGT-win/CatGT.exe -dir=E:\Task2-SWR\M590\rawdata -run=M590-2025-03-31 -g=0 -t=0 -prb_fld -t_miss_ok -ni -maxsecs=5761 -xa='0,0,0,1,3,0' -xia='0,0,0,1,3,0' -xa='0,0,1,1,3,0' -xia='0,0,1,1,3,0' -xa='0,0,2,1,3,0' -xia='0,0,2,1,3,0' -xa='0,0,3,1,3,0' -xia='0,0,3,1,3,0' -xa='0,0,4,1,3,0' -xia='0,0,4,1,3,0' -xa='0,0,5,1,3,0' -xia='0,0,5,1,3,0' -xa='0,0,6,1,3,0' -xia='0,0,6,1,3,0' -xa='0,0,7,1,3,0' -xia='0,0,7,1,3,0' -xd='0,0,8,2,0' -xid='0,0,8,2,0' -xd='0,0,8,3,0' -xid='0,0,8,3,0' -dest=D:\Test\
```